

Experiment 1A

Aim

To study the fundamental elements of Computer Networking, including Network Topologies, Transmission Media, Networking Devices, and the RJ-45 connector along with its color coding schemes.

Theory

A. Network Topologies

A network topology defines how computers, cables, and other components are arranged and connected in a network.

1. Bus Topology

- **Description:** All nodes are connected to a single central cable, called the backbone.
- **Advantages:** Easy to set up and requires minimal cabling.
- **Disadvantages:** If the central cable fails, the entire network goes down.

2. Star Topology

- **Description:** All devices are individually connected to a central networking device (Hub or Switch).
- **Advantages:** Easy to troubleshoot; failure of one cable does not affect other devices.
- **Disadvantages:** If the central hub/switch fails, the whole network becomes unreachable.

3. Ring Topology

- **Description:** Each device is connected to two neighboring devices, forming a circular data path.
- **Advantages:** Equal access to resources; no collisions as data moves in one direction.
- **Disadvantages:** A single broken connection disables the entire network.

4. Mesh Topology

- **Description:** Every device has a point-to-point connection to every other device in the network.
- **Advantages:** Highly reliable, secure, and offers dedicated links with no traffic congestion.
- **Disadvantages:** High cabling cost and complex installation.

5. Tree Topology

- **Description:** A hierarchical combination of Star and Bus topologies, organized in groups like a tree structure.
- **Advantages:** Highly scalable and easy to manage in large organizations.
- **Disadvantages:** Heavy reliance on the main backbone cable.

6. Hybrid Topology

- **Description:** A combination of two or more different network topologies (e.g., Star-Ring).
- **Advantages:** Flexible and scalable to match specific organizational needs.
- **Disadvantages:** Complex structure and higher maintenance costs.

B. Transmission Media

Transmission media carry information from a transmitter to a receiver across a network using physical or guided paths.

1. Coaxial Cable :

- Solid copper conductor wrapped in insulation, a braided metal shield, and an outer jacket.

- Moderate bandwidth, resistant to electrical interference.
- Cable TV networks and older LAN setups.

2. **Twisted Pair Cable**

- Pairs of insulated copper wires twisted together to reduce electromagnetic interference.
- Available in Shielded (STP) and Unshielded (UTP) variants; low cost.
- Standard Ethernet local area networks (LANs).

3. **Optical Fiber**

- Glass or plastic core that transmits data as light pulses.
- High bandwidth, extremely fast, completely immune to EMI.
- Internet backbones and long-distance telecom.

C. **Networking Devices**

- **Router:** An intelligent device working at the Network Layer (Layer 3) that routes data packets across different networks using IP addresses.
- **Hub:** A basic Physical Layer (Layer 1) device that broadcasts all incoming data packets to every connected port, causing network collisions.
- **Switch:** A Data Link Layer (Layer 2) device that forwards incoming frames directly to the intended target device using MAC addresses.
- **Gateway:** A system or device acting as a translation bridge between two networks running completely different communication protocols.
- **Modem (Modulator-Demodulator):** Converts digital signals from computers into analog signals for transmission over phone/cable lines, and vice versa.
- **NIC (Network Interface Card):** A hardware component (PCI card or integrated chip) that provides a physical MAC address interface to connect a computer to a network.

D. **RJ-45 Connector & Color Coding**

An **RJ-45 (Registered Jack 45)** is an 8-pin connector used to terminate Ethernet cables (like Cat5e and Cat6).

Color Coding Schemes (TIA/EIA-568)

1. **T568A Standard:**

- Pin 1: White / Green
- Pin 2: Green
- Pin 3: White / Orange
- Pin 4: Blue
- Pin 5: White / Blue
- Pin 6: Orange
- Pin 7: White / Brown
- Pin 8: Brown

2. **T568B Standard:**

- Pin 1: White / Orange
- Pin 2: Orange
- Pin 3: White / Green
- Pin 4: Blue
- Pin 5: White / Blue
- Pin 6: Green

- Pin 7: White / Brown
- Pin 8: Brown

Conclusion

Through this study, the fundamental components of computer networking, including topologies, transmission media, layer-specific hardware devices, and RJ-45 cabling specifications, were analysed and documented.